Compound angle equations

1

a $\alpha = 45^\circ, 315^\circ$

b $\theta = 25^\circ, 295^\circ$

2

a $\alpha = 60^\circ, -60^\circ$

b $\theta = 160^\circ, 40^\circ$

3

a $\alpha = 390^\circ, 150^\circ$

b $\theta = 355^\circ, 115^\circ$

4

a $\alpha = 60^\circ, 120^\circ$

b $\theta = 35^\circ, 95^\circ$

5

a $\alpha = 30^\circ, 210^\circ$

b $\theta = 15^\circ, 195^\circ$

6

a $\theta = 300^\circ$

b $\theta = 210^\circ, 300^\circ$

c $\theta = 195^\circ, 255^\circ$

d $\theta = 75^\circ, 255^\circ$

e $\theta = 90^\circ, 270^\circ$

f $\theta = 0^\circ, 300^\circ, 360^\circ$

7

a $\theta = 15^\circ, 105^\circ, 135^\circ, 225^\circ, 255^\circ, 345^\circ$

b $\theta = 105^\circ, 165^\circ, 285^\circ, 345^\circ$

c $\theta = 22.79^\circ, 157.21^\circ, 202.79^\circ, 337.21^\circ$

d $\theta = 15^\circ, 45^\circ, 135^\circ, 165^\circ, 255^\circ, 285^\circ$

e $\theta = 120^\circ, 240^\circ$

f $\theta = 45^\circ, 105^\circ, 165^\circ, 225^\circ, 285^\circ, 345^\circ$

8

a $x = 15^\circ, 60^\circ, 105^\circ, 150^\circ$

b $x = 30^\circ, 60^\circ$

c $x = 45^\circ, 75^\circ, 165^\circ$

d $x = 135^\circ$

9

a $x = -165^\circ, -75^\circ, -45^\circ, 45^\circ, 75^\circ, 165^\circ$

b $x = -165^\circ, -105^\circ, 15^\circ, 75^\circ$

c $x = \pm 120^\circ$

d $x = -90^\circ, 30^\circ, 150^\circ$

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Squared equations

10

a $\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$

b $\theta = 60^\circ, 120^\circ, 240^\circ, 300^\circ$

c $\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$

d $\theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$

e $\theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ$

f $\theta = 0^\circ, 360^\circ$

g $\theta = 22.5^\circ, 67.5^\circ, 112.5^\circ, 157.5^\circ, 202.5^\circ, 247.5^\circ, 292.5^\circ, 337.5^\circ$

h $\theta = 10^\circ, 50^\circ, 70^\circ, 110^\circ, 130^\circ, 170^\circ, 190^\circ, 230^\circ, 250^\circ, 290^\circ, 310^\circ, 350^\circ$

11

a $\theta = 30^\circ$

b $\theta = 120^\circ, 240^\circ$

Factorisation

12 Zero

13

a $\theta = 45^\circ, 225^\circ$

b $\theta = 30^\circ, 90^\circ, 150^\circ, 210^\circ, 270^\circ, 330^\circ$

c $\theta = 22^\circ, 117^\circ, 202^\circ, 297^\circ$

d $\theta = 61.7^\circ, 154.6^\circ, 205.4^\circ, 298.3^\circ$

e $\theta = 21.3^\circ, 158.7^\circ, 246.7^\circ, 293.3^\circ$

f $\theta = 90^\circ, 270^\circ, 45^\circ, 225^\circ$

g $\theta = 0^\circ, 180^\circ, 360^\circ, 60^\circ, 300^\circ$

14

a $\theta = 60^\circ, 45^\circ$

b $\theta = 30^\circ$

15

a $\theta = 210^\circ, 225^\circ$

b $\theta = 315^\circ$

c $\theta = 0^\circ, 30^\circ$

d $\theta = -180^\circ, 0^\circ, 30^\circ, 150^\circ, 180^\circ$

16 $\sin\theta + 1 = 0$

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Equations and trigonometric identities

17 $\theta = 60^\circ$

18

a $\theta = 0^\circ, 30^\circ, 150^\circ, 180^\circ$

b $\theta = 0^\circ, 90^\circ, 180^\circ, 270^\circ, 360^\circ$

c $\theta = 34^\circ, 63^\circ, 214^\circ, 243^\circ$

d $\theta = 60^\circ, 180^\circ, 300^\circ$

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